

REFERENCE SHEET: THE MAHLEREXPERIMENTS FOLDER

Layout. Each experiment family (`rk`, `v2`, `v3`, `v5`, `v7`) has one subfolder per feature combination, named as in Table 1. Inside an experiment folder:

- `data/` — the training/validation data used for that experiment;
- one subfolder per training run (e.g. `v2_fD run 3 (best run)`); for the `rk` experiments the runs are grouped into `exact/` and `size/` according to the accuracy criterion used (see below).

A run folder contains the training log (`train.log`), the model parameters (`params.pkl`), the final checkpoint (`checkpoint.pth`), and the per-epoch validation dumps `eval.valid.elliptic_curve.<epoch>`. The run marked (*best run*) is the one used for the figures in the paper. The top-level file `data/output_full.txt` is the master data file: for each k it stores r_k (as a fraction), the curve E_k with its conductor N_k and further invariants; the plotting scripts read $n_k = 1/r_k$, N_k , etc. from it.

Accuracy criteria (exact vs. size). The `rk` experiments learn r_k as $[\text{sign}, |n_k|]$ with $|n_k| = 1/|r_k|$. Two criteria are used:

- *exact*: the model output must match the answer $[\text{sign}, |n_k|]$ exactly;
- *size*: an output s_k is counted correct iff it has the same sign as r_k and $|r_k|/2 < |s_k| < 2|r_k|$ (the factor-2 window; since $s \leftrightarrow 1/s$ preserves the window, this is the test $|\log(\text{guess}) - \log(\text{true})| < \log 2$ on the stored magnitude $|n_k|$).

For the v_p experiments the target $v_p(r_k)$ is a small integer and plain exact match is used.

Notation. ω = root number ω_k ; N = conductor N_k ; Δ = discriminant Δ_k ; a_p = Frobenius traces; $c_2 = \dots$; rank = Mordell–Weil rank; `factor(·)` = the quantity is given to the model in factored form. All experiments receive k as input; the extra features listed in Table 1 are given in addition.

TABLE 1. Explanation of the folder names.

Experiment	Target	Extra features
rk		none
rk_w		ω
rk_wN		ω, N
rk_wap		ω, a_p
rk_wNap	r_k	ω, N, a_p
rk_wfN		$\omega, \text{factor}(N)$
rk_wD		ω, Δ
rk_wfD		$\omega, \text{factor}(\Delta)$
rk_wr		ω, rank
v2		none
v2_N		N
v2_fN		$\text{factor}(N)$
v2_D	$v_2(r_k)$	Δ
v2_fD		$\text{factor}(\Delta)$
v2_fDfN		$\text{factor}(\Delta), \text{factor}(N)$
v2_c2		c_2
v2_fDc2		$\text{factor}(\Delta), c_2$
v3		none
v3_N		N
v3_fN	$v_3(r_k)$	$\text{factor}(N)$
v3_D		Δ
v3_fD		$\text{factor}(\Delta)$
v3_fDfN		$\text{factor}(\Delta), \text{factor}(N)$
v5		none
v5_fN	$v_5(r_k)$	$\text{factor}(N)$
v5_fD		$\text{factor}(\Delta)$
v7		none
v7_fN	$v_7(r_k)$	$\text{factor}(N)$

TABLE 2. Runs used for the figures.

Figure(s)	Run folder
3.1, 3.3, 3.4, 6.1, 6.2	data/output_full.txt
5.1	rk/rk/size/rk run 2 (size) (best run) rk/rk_w/size/rk_w (size)
5.2	rk/rk_wN/final-03e.PB3-exp rk/rk_wfN/size/rk_wfN (size) rk/rk_wfD/size/rk_wfD (size)
5.3	rk/rk_wfD/size/rk_wfD (size)
5.4–5.8	v3/v3_fD/v3_fD run 2 (best run)
5.9–5.13	v2/v2_fD/v2_fD run 3 (best run)